

# Temporal Robustness of Liquid Neural Networks under Irregular Observation Dropout

A Controlled Null Result on Satellite Image Time Series

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**Abstract.** Closed-form continuous-time (CfC) networks embed the true, irregular inter-observation interval  $\Delta t$  directly in their ODE dynamics, which suggests they should tolerate missing observations more gracefully than discrete recurrent baselines. We test this hypothesis directly on Sentinel-2 image time series (PASTIS), on a phenologically confusable crop pair (*Soft winter wheat* vs. *Corn*) whose separation depends on temporal trajectory rather than any single frame. Using an *honest twin* design—a shared CNN encoder with only the recurrent core swapped between CfC and GRU—and a pre-registered, frozen decision criterion, we measure classification retention as increasing fractions of satellite revisits are removed at test time. The pre-registered verdict is a **null**: the CfC retention advantage at 60% dropout (+0.113) does not exceed the pooled standard deviation (0.176). We show this null reflects *overlapping distributions with high population spread*, not absence of effect: the directional signal is consistent across three independent measures—CfC degrades with a shallower slope (−0.51 vs. −0.69) and, at heavy dropout, overtakes GRU in absolute  $F_1$  (0.564 vs. 0.480) despite starting lower. A pre-registered increase of the seed count from 3 to 15 did not shrink the variance, confirming the effect is genuinely below the significance threshold on this task rather than merely under-powered. We draw three methodological conclusions and argue that the open-loop classification setting structurally dampens the temporal mechanism, motivating a move to closed-loop control tasks where liquid networks have documented out-of-distribution advantages.

## 1 Motivation

Liquid neural networks—Liquid Time-constant networks and their closed-form successor (CfC) [1, 2]—model a hidden state as a continuous-time dynamical system. A practically important property is that the elapsed physical time between two inputs,  $\Delta t$ , enters the state update *natively* through the ODE, rather than as an appended feature. On paper this makes them well suited to sensing regimes where observations arrive at irregular, unpredictable intervals.

On-board Earth-observation (EO) triage is exactly such a regime. A satellite revisits a tile at intervals that vary with orbit and are frequently broken by cloud cover, so the usable time series has irregular gaps. If a compact recurrent core degrades gracefully as observations are lost, it becomes a candidate component for on-board decision-making under bandwidth and power constraints. This report asks a single, falsifiable question:

*Does a CfC core, carrying the true  $\Delta t$  of surviving observations, degrade more slowly than a size-matched GRU baseline as satellite revisits are removed?*

## 2 Task and Data

**Substrate.** PASTIS [3], real Sentinel-2 image time series over France with crop labels. We deliberately reduce the problem to a *binary* classification of a phenologically confusable pair, *Soft winter wheat* vs. *Corn*. These two classes are difficult to separate from any single acquisition but follow different growth calendars, so their discrimination depends on the *temporal trajectory*—precisely the regime in which a temporal mechanism can express an advantage. This choice was made after a diagnostic (Section 6) showed that aggregating a dominant crop class over a whole tile suppresses temporal information.

**Split.** 626 patches, partitioned by PASTIS Fold: train  $\{1, 2, 3\}$ , validation  $\{4\}$ , test  $\{5\}$  (spatially disjoint, no leakage).

**OOD axis (temporal, primary).** At test time only, a fraction  $d \in \{0.0, 0.2, 0.4, 0.6\}$  of acquisitions is removed at random, simulating cloud gaps;  $\Delta t$  is recomputed from the dates of the *surviving* acquisitions, with a floor of two revisits. Training always uses the full cadence ( $d = 0$ ). The question is thus about generalization to sparser observation, not about training on degraded data.

### 3 Method

**Honest twin.** Both models share an identical CNN encoder (28,752 parameters); they differ *only* in the recurrent core (CfC vs. GRU). The interval tensor  $\Delta t$  is passed symmetrically to both (GRU ignores it internally). Per-model learning rates were fixed by a separate, symmetric sweep prior to the main experiment (CfC:  $3 \times 10^{-4}$ ; GRU:  $1 \times 10^{-3}$ ); no other hyperparameter differs between twins. This isolates the effect of the recurrent mechanism from encoder capacity and tuning budget.

**Frozen pre-registration.** The decision criterion was written to disk *before* the run and left untouched afterwards:

- **Primary metric**—retention relative to each model’s own baseline,  $R_{\text{model}}(d) = F_1(d)/F_1(0)$ . Using retention (rather than absolute  $F_1$ ) neutralizes any starting-point handicap and asks how much *each* model loses from *its own* best.
- **Primary indicator**— $R(0.6)$ , retention at the heaviest dropout.
- **Threshold**—the thesis is confirmed *iff* the margin  $R_{\text{CfC}}(0.6) - R_{\text{GRU}}(0.6)$  exceeds the pooled standard deviation.
- **Variance**—15 training seeds  $\times$  10 random dropout masks = 150 measurements per point; report mean  $\pm$  standard deviation.

**Amendment (also pre-registered).** After an initial  $n = 3$  run, the seed count was raised to  $n = 15$  for *statistical power only*; metric, indicator, threshold, OOD axis, task, and learning rates were unchanged. A finality rule was recorded to prevent optional stopping: the  $n = 15$  result is binding regardless of outcome, and no further increase of  $n$  is permitted after seeing it.

### 4 Results

**Full-cadence reference** ( $d = 0$ , macro- $F_1$  on test): CfC  $0.8802 \pm 0.0121$  versus GRU  $0.9073 \pm 0.0137$ . GRU starts approximately 0.027 higher—a starting-point handicap for CfC.

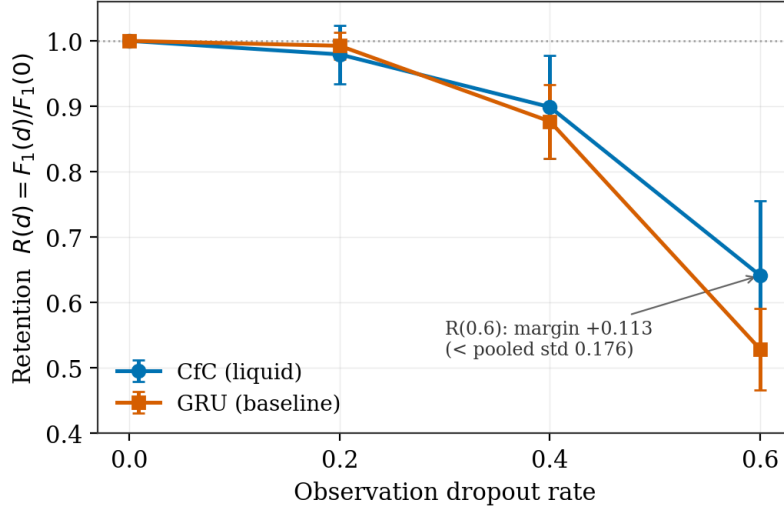
**Table 1:** Retention  $R(d) = F_1(d)/F_1(0)$ ,  $n = 15$  (150 measurements per cell).

| Dropout $d$ | $R_{\text{CfC}}$                      | $R_{\text{GRU}}$                      |
|-------------|---------------------------------------|---------------------------------------|
| 0.0         | $1.0000 \pm 0.0000$                   | $1.0000 \pm 0.0000$                   |
| 0.2         | $0.9793 \pm 0.0445$                   | $0.9925 \pm 0.0202$                   |
| 0.4         | $0.8989 \pm 0.0785$                   | $0.8770 \pm 0.0565$                   |
| <b>0.6</b>  | <b><math>0.6415 \pm 0.1138</math></b> | <b><math>0.5288 \pm 0.0622</math></b> |

**Primary indicator.** At  $d = 0.6$  the margin is  $+0.1127$  against a pooled standard deviation of 0.1759. The margin does *not* exceed the pooled standard deviation.

**Verdict: thesis *not* confirmed** under the frozen criterion.

Linear-fit slope of  $F_1$  against dropout: CfC  $-0.5095$  versus GRU  $-0.6936$ .



**Figure 1:** Retention  $R(d)$  with  $\pm$ std error bars. The CfC advantage at heavy dropout is real in direction but sits within the pooled spread.

**Table 2:** Context—absolute macro- $F_1$  by dropout rate,  $n = 15$ .

| Dropout $d$ | $F_1$ CfC                             | $F_1$ GRU                             |
|-------------|---------------------------------------|---------------------------------------|
| 0.0         | $0.8802 \pm 0.0121$                   | $0.9073 \pm 0.0137$                   |
| 0.2         | $0.8621 \pm 0.0424$                   | $0.9004 \pm 0.0211$                   |
| 0.4         | $0.7911 \pm 0.0691$                   | $0.7959 \pm 0.0548$                   |
| 0.6         | <b><math>0.5642 \pm 0.0985</math></b> | <b><math>0.4798 \pm 0.0567</math></b> |

## 5 Analysis: a null is not an absence of effect

The null verdict reflects *overlapping distributions with high population spread*, not the absence of a mechanism. The directional signal is consistent across three independent measures:

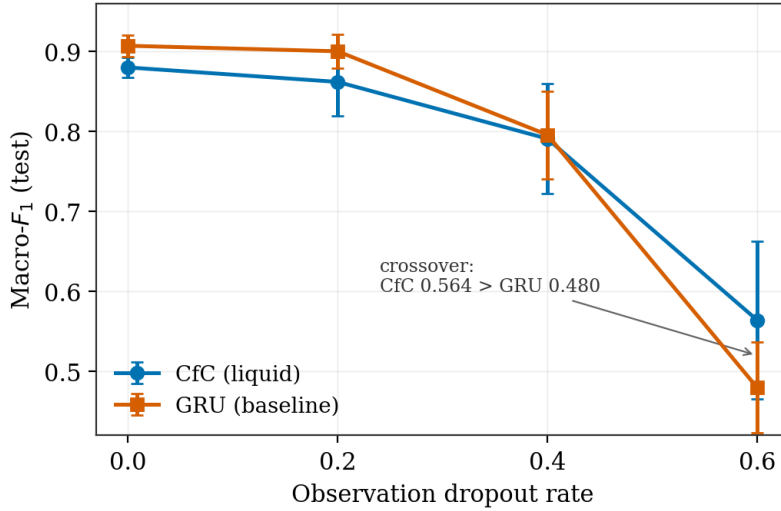
1. retention  $R_{\text{CfC}} \geq R_{\text{GRU}}$  at  $d = 0.4$  (0.899 vs. 0.877) and  $d = 0.6$  (0.642 vs. 0.529);
2. at  $d = 0.6$ , CfC *overtakes* GRU in absolute  $F_1$  (0.564 vs. 0.480) despite a 0.027 lower start—the curves cross;
3. the degradation slope of CfC is shallower ( $-0.51$  vs.  $-0.69$ ).

**The “under-powered” hypothesis was tested and rejected.** Raising the seed count from 3 to 15 did *not* shrink the spread:

**Table 3:** Effect of increasing seeds on the primary indicator.

|          | margin $R(0.6)$ | std CfC | std GRU | pooled std |
|----------|-----------------|---------|---------|------------|
| $n = 3$  | +0.1099         | 0.1170  | 0.0537  | 0.1706     |
| $n = 15$ | +0.1127         | 0.1138  | 0.0622  | 0.1759     |

The standard deviation reported here is the *population spread* of retention across masks and seeds, not the standard error of the mean; increasing the number of seeds sharpens the estimate of the mean but does not reduce the underlying spread. Margin ( $\approx 0.11$ ) and spread ( $\approx 0.18$ ) are stable across the fivefold increase in  $n$ . The mechanism therefore acts in the predicted direction, but its effect does not clear the pre-registered significance threshold on this task at this level of variance.



**Figure 2:** Absolute macro- $F_1$ . The curves *cross*: GRU leads at full cadence, but at  $d = 0.6$  CfC overtakes it (0.564 vs. 0.480) despite starting lower.

## 6 Methodological conclusions

- (a) **Metric power.** Retention, being a ratio, carries high variance and low power to detect this effect; the absolute slope reveals it more cleanly ( $-0.51$  vs.  $-0.69$ ). The primary-verdict metric was conservative. Crucially, because it was frozen before the run, we do *not* switch the verdict to the slope—the directional slope is reported as a secondary signal, not as the headline result. Retaining this discipline is what keeps the null honest.
- (b) **Task design governs whether the mechanism can show.** On the full four-class dominant-crop task the CfC–GRU gap was 0.072 (structural); on the trajectory-dependent pair it narrowed to 0.028. Aggregating a dominant class over a tile averages away the temporal structure the CfC core exists to exploit. The choice of target, not the choice of architecture, determined whether a temporal advantage was even observable.
- (c) **Regime matters.** At full cadence GRU is simply better (0.907 vs. 0.880). The CfC advantage, where it exists, lives in the *sparse-observation regime* (high dropout), which is exactly where the absolute curves cross.

## 7 Conclusion and next step

On an open-loop classification task, with real satellite time series and a rigorously pre-registered protocol, the temporal advantage of a native- $\Delta t$  liquid core over a discrete baseline is *real in direction but subtle in magnitude*—consistent across three measures, yet below a pre-registered significance threshold that fifteen seeds could not move. This is a clean, reproducible negative result, and it is informative: it locates the CfC advantage in the sparse-observation regime and shows that open-loop classification structurally dampens it.

This motivates a change of setting rather than a change of architecture. Liquid networks have their strongest documented advantage in *closed-loop control*, where a model’s decision changes its next observation and irregular, out-of-distribution dynamics are the norm—most notably in robust out-of-distribution flight navigation [4]. The natural next step is to move the same honest-twin, frozen-criterion methodology used here onto a closed-loop navigation task under distribution shift, where the temporal mechanism has room to express the advantage that open-loop classification suppresses. The contribution of this phase is therefore twofold: a documented null that maps the boundary of the mechanism, and a validated experimental harness—honest twin, pre-registered criterion, dropout-and-variance protocol—ready to transfer to that setting.

## Reproducibility

Python 3.10; torch==2.12.1+cu130 (GPU: GTX 1660 Ti) or CPU build 2.12.1+cpu; ncps==1.0.1. Training seeds 0–14; dropout-mask seeds 0–9. The recurrent core implementation is device-agnostic and was left unmodified throughout the project. Frozen criterion, binding  $n = 15$  results, the  $n = 3$  run, and both figures are retained as artifacts.

## References

- [1] R. Hasani, M. Lechner, A. Amini, D. Rus, R. Grosu. “Liquid Time-constant Networks.” *AAAI*, 2021.
- [2] R. Hasani, M. Lechner, A. Amini, et al. “Closed-form continuous-time neural networks.” *Nature Machine Intelligence*, 2022.
- [3] V. Sainte Fare Garnot, L. Landrieu. “Panoptic Segmentation of Satellite Image Time Series with Convolutional Temporal Attention Networks.” *ICCV*, 2021.
- [4] M. Chahine, R. Hasani, et al. “Robust flight navigation out of distribution with liquid neural networks.” *Science Robotics*, 2023.